

Tactical Role Clustering in the Premier League Using Unsupervised Machine Learning

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1 Introduction and Motivation

Traditional soccer positions such as defender, midfielder, and forward are useful, but they do not always describe how a player actually contributes during a match. Modern Premier League players often perform hybrid roles. Some forwards contribute heavily to buildup, some midfielders play like attackers, and some defenders are important ball progressors. Because of this, I wanted to investigate whether statistical player profiles could reveal tactical roles beyond the listed position labels.

The main research question for this project is: *Can Premier League players be grouped into meaningful tactical roles using attacking and ball-progression statistics rather than only traditional position labels?* To answer this question, I used unsupervised machine learning to cluster players based on per-90 performance metrics. Instead of predicting a target variable, the goal was to discover natural groupings in the data.

The dataset used in this project is the FBref Premier League 2024/25 player statistics dataset from Kaggle, compiled by Siddhraj Thakor. The raw dataset contains 574 player rows and 36 columns. Each row represents a player's season-level performance, including information such as team, position, minutes played, goals, assists, expected goals, expected assisted goals, progressive carries, progressive passes, and progressive receptions. These features are useful for comparing player roles because they capture both attacking output and ball progression.

2 Data Description and Cleaning

The dataset was loaded into Python using pandas. I first inspected the dataset shape, column names, data types, missing values, and numerical summary statistics. The raw dataset contained 574 players and 36 columns. Some columns were categorical, such as player name, nationality, squad, and position. Other columns were numerical, such as minutes played, goals, assists, expected goals, and progression statistics.

A major issue with this type of soccer data is that players do not all play the same number of minutes. A player with very limited minutes can have misleading per-90 statistics because one or two events can greatly affect the rate. To reduce this problem, I filtered the dataset to include only players with at least 600 minutes played. I used a 600-minute cutoff because it removes very small samples while still keeping rotation players in the dataset. A stricter cutoff, such as 900 minutes, would reduce noise further, but it would also remove many useful player examples. After filtering, the dataset was reduced from 574 players to 369 players.

The project focuses on per-90 statistics because they allow players with different playing time to be compared more fairly. Some attacking columns were already available as per-90 values, while the raw progression columns needed to be converted. I created new features for progressive carries per 90, progressive passes per 90, and progressive receptions per 90 by dividing each total by the player's number of 90-minute matches.

The final clustering features were:

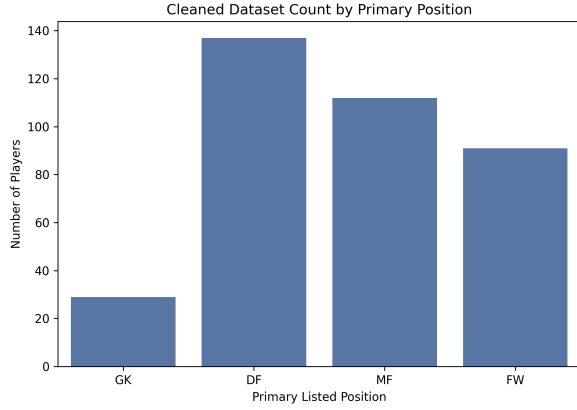
- Goals per 90
- Assists per 90
- Non-penalty expected goals per 90
- Expected assisted goals per 90
- Progressive carries per 90
- Progressive passes per 90
- Progressive receptions per 90

I used non-penalty expected goals instead of total expected goals because penalties can inflate attacking output and may not reflect open-play tactical role as clearly. Before clustering, the selected features were standardized using `StandardScaler`. This was necessary because K-Means and other distance-based clustering methods can be affected by feature scale.

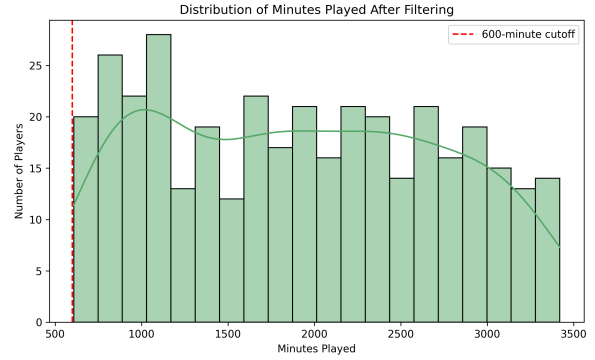
3 Exploratory Analysis

Before applying clustering, I explored the cleaned dataset to understand its structure. I looked at the distribution of player positions, the distribution of minutes played, and the relationships among the selected clustering features.

Figure 1 shows that the cleaned dataset still contains players from all major listed positions: goalkeepers, defenders, midfielders, and forwards. Defenders are the largest group in the cleaned data, while goalkeepers are the smallest. The minutes distribution also shows that, even after filtering, players still range from just above 600 minutes to more than 3000 minutes. This supports the decision to use per-90 statistics instead of raw season totals because regular starters would otherwise dominate the results simply because they played more minutes.



(a) Count of players by primary listed position after the 600-minute filter. Defenders make up the largest group, followed by midfielders and forwards. Goalkeepers are the smallest group.



(b) Distribution of minutes played after filtering. The red dashed line shows the 600-minute cutoff. Players still vary widely in playing time, which supports using per-90 statistics.

Figure 1: Basic exploratory plots for the cleaned dataset.

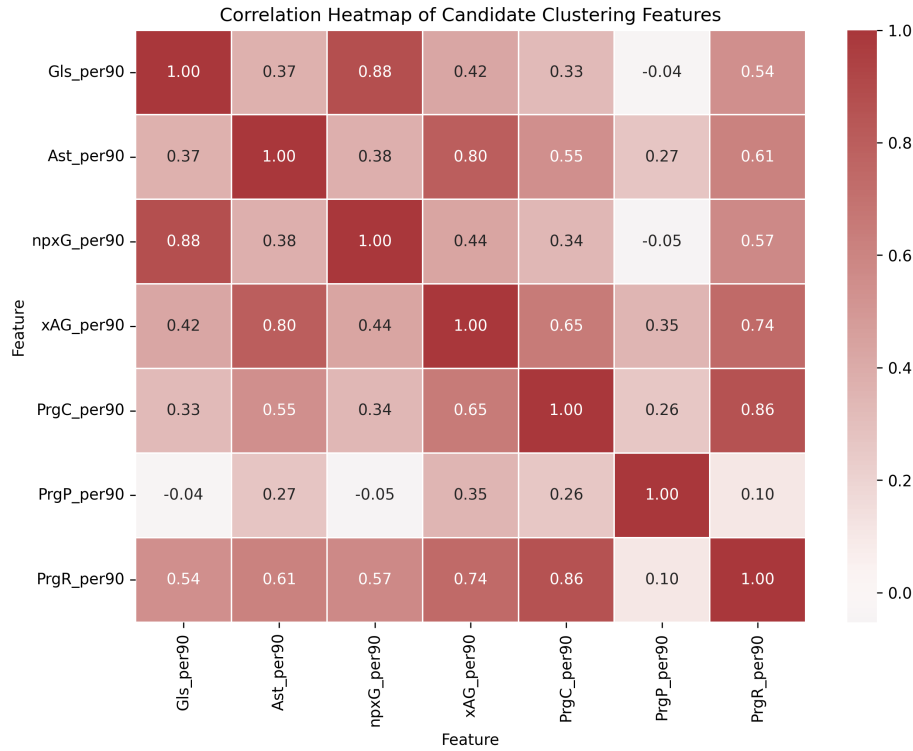


Figure 2: Correlation heatmap of candidate clustering features. Goals per 90 and non-penalty expected goals per 90 are strongly related, while progressive passes per 90 has weaker relationships with the scoring features.

Figure 2 shows several expected relationships among the selected features. Goals per 90 and non-penalty expected goals per 90 have a strong positive correlation, which suggests that the scoring features measure a similar type of attacking threat. Assists per 90 and expected

assisted goals per 90 are also strongly related. The progression features are not completely separate from attacking output: progressive carries and progressive receptions are highly correlated, and expected assisted goals is also related to progressive receptions. Progressive passes per 90 has weaker relationships with the scoring features, suggesting that it captures a different type of ball-progression contribution.

Overall, the exploratory analysis showed that the dataset contains a wide range of positions and playing-time levels. It also showed that the selected features contain related but not identical information, which makes them reasonable inputs for clustering.

4 Methods and Results

4.1 K-Means Clustering

The main machine learning technique used in this project was K-Means clustering. K-Means is appropriate for this research question because there is no existing target label for tactical role. The goal is to group players based on similarity in their statistical profiles.

I evaluated cluster counts from $k = 2$ to $k = 10$ using inertia and silhouette score. Inertia measures how compact the clusters are, while the silhouette score measures how well-separated the clusters are. A higher silhouette score usually indicates better-defined clusters.

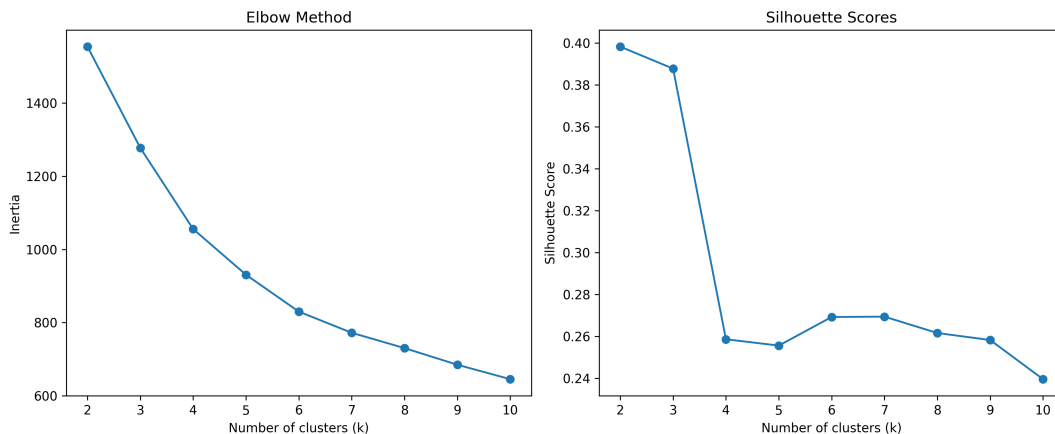


Figure 3: K-Means evaluation using the elbow method and silhouette scores. The silhouette score is highest for $k = 2$, but $k = 3$ provides a more useful tactical interpretation.

As shown in Figure 3, $k = 2$ had the highest silhouette score, but it mostly separated players into a broad low-attacking group and a broad attacking group. That split was statistically clean, but it was not very useful for the research question. I selected $k = 3$ because it still had a similar silhouette score and created a more meaningful separation between goal-scoring forwards and creative or progressive attackers.

The final K-Means model used $k = 3$, `random_state=42`, and `n_init=10` for reproducibility. The three clusters were interpreted as:

- Defensive / Low-Attacking Involvement
- Goal-Scoring Forwards
- Creative / Progressive Hybrid Attackers

These names were assigned after reviewing the average feature values for each cluster and example players from each group.

4.2 PCA Visualization

To visualize the clusters, I used Principal Component Analysis (PCA). PCA reduces the seven-dimensional feature space into two principal components so the clusters can be plotted in two dimensions. The first two principal components explained about 76% of the variation in the selected features, which makes the two-dimensional visualization useful for interpretation.

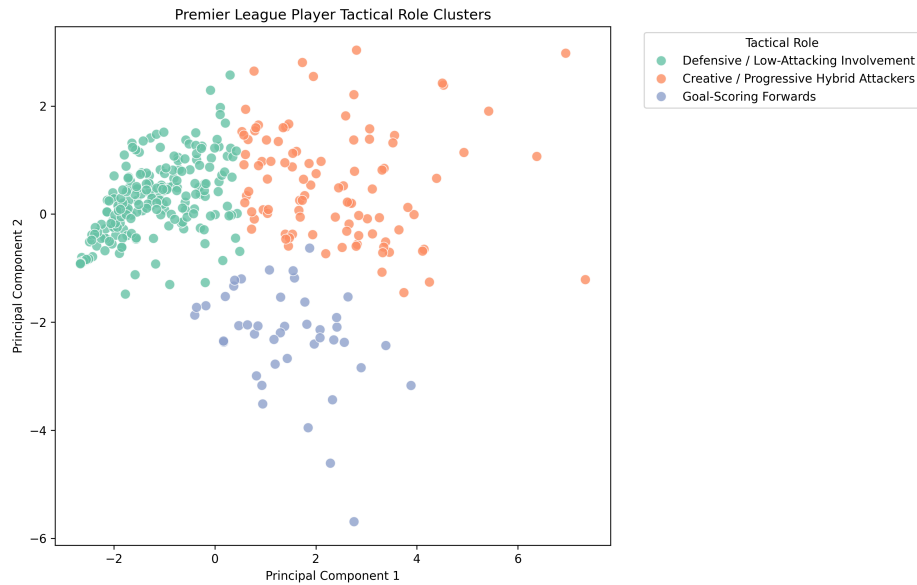


Figure 4: PCA visualization of Premier League players colored by assigned tactical role. The three clusters show broad separation based on attacking and progression profiles.

Figure 4 shows that the defensive or low-attacking group is mostly separated from the two more attacking groups. The creative/progressive hybrid attackers are generally farther toward the upper and right side of the plot, while the goal-scoring forwards form a more distinct group lower in the PCA space. This supports the idea that the model is identifying broad attacking and progression-based roles.

4.3 Cluster Profiles

The average per-90 statistics for each tactical role are shown in Figure 5. This heatmap was the main tool used to interpret the clusters.



Figure 5: Average per-90 statistics by tactical role. Goal-scoring forwards have the highest scoring-related values, while creative/progressive hybrid attackers have the highest creative and ball-progression values.

The goal-scoring forward cluster has the highest goals per 90 and non-penalty expected goals per 90. This suggests that these players are mainly separated by direct scoring threat and shot quality. The creative/progressive hybrid attacker cluster has the highest assists per 90, expected assisted goals per 90, progressive carries per 90, progressive passes per 90, and progressive receptions per 90. This group represents players who contribute more through creation, movement, and ball progression. The defensive/low-attacking involvement group has much lower values for scoring, expected goals, assists, expected assisted goals, carries, and receptions. However, this group still has some progressive passing value, which is reasonable because defenders and deeper midfielders can still progress the ball through passing.

One important observation is that all high-minute goalkeepers naturally fall into the defensive/low-attacking involvement group. This is expected because the selected features are focused on attacking and progression rather than goalkeeping actions.

4.4 CS590 Additional Analysis: Clustering Comparison

For the graduate-level part of the project, I compared K-Means with Agglomerative Clustering using the same scaled feature matrix and the same number of clusters. This comparison was included to check whether the tactical groupings were stable across more than one unsupervised learning method.

Table 1: Model comparison for three-cluster solutions.

Model	Number of Clusters	Silhouette Score
K-Means	3	0.388
Agglomerative Clustering	3	0.205

As shown in Table 1, K-Means produced a higher silhouette score than Agglomerative Clustering for the selected three-cluster structure. I also calculated the Adjusted Rand Index between the two sets of cluster assignments. The ARI was about 0.245, which suggests that the two methods agreed only partially. This means that some player assignments are sensitive to the clustering method used. That is a useful limitation to acknowledge, but K-Means remained the main model because it produced a higher silhouette score and more interpretable tactical groupings.

5 Discussion and Conclusions

The results suggest that unsupervised learning can reveal useful attacking and progression-based player profiles that do not always match traditional position labels. The final K-Means model separated players into three broad tactical groups: defensive or low-attacking involvement players, goal-scoring forwards, and creative/progressive hybrid attackers.

The most important finding is that the model separated different types of attacking contribution. Goal-scoring forwards were defined by high goals per 90 and non-penalty expected goals per 90. Creative/progressive hybrid attackers were defined more by assists, expected assisted goals, progressive carries, progressive passes, and progressive receptions. This supports the idea that modern attacking roles are not only about scoring goals. Some players contribute more by carrying the ball, receiving progressive passes, and creating chances for others.

The analysis also shows why listed position alone can be limited. A forward, midfielder, or wide player may appear in a similar tactical cluster if their statistical profiles are similar. At the same time, the project does not claim to capture every tactical role in soccer. The feature set is focused on attacking and ball progression, so the model is better at separating attacking roles than distinguishing defensive roles such as center-backs, defensive midfielders, and goalkeepers.

There are several limitations. First, the dataset only covers one Premier League season, so the clusters may change if multiple seasons are included. Second, the model uses a limited set of features and does not include detailed defensive, pressing, passing-network, or goalkeeper-specific metrics. Third, the choice of $k = 3$ involves interpretation as well as metrics. Although $k = 2$ had the highest silhouette score, $k = 3$ was selected because it gave a more useful tactical answer to the research question. Finally, the comparison with Agglomerative Clustering showed that some assignments are sensitive to the algorithm used.

With more time, I would expand the project by adding defensive and possession metrics,

comparing multiple seasons, and separating out goalkeepers before clustering. I would also test other unsupervised methods such as DBSCAN or Gaussian Mixture Models. Another useful extension would be to compare the clusters with expert-defined player roles or team tactical systems.

Overall, the project answers the research question by showing that statistical clustering can be a useful exploratory tool for identifying broad attacking and progression-based tactical patterns in Premier League player data.